R46: exact polynomial-root collection

Fixed-chart observations, not an independent verdict or IC speed comparison.

Detector inputs; all complete point sets match

Degree	R45 lambda trials	R46 algebraic candidates	
7	17,920	420	byte-identical
9	276,480	540	byte-identical
11	4,055,040	2,860	byte-identical
13	66,027,520	7,540	byte-identical

control7/9/11/13-01; different stages, not total cost or IC performance

Exact substitution and complete roots

The registered equation gives physical nu(lambda)=N(lambda)/C. Its gamma coefficient is checked identically equal to b. Substitute into every source equation and take the common GCD of both coefficient components. Intersect with lambda raised to q minus lambda, then split with field-basis trace polynomials. Divisions and final linear factors reconstruct exactly; every root re-evaluates in the original GCD. All original equations, eight units, coefficient descent, Tate order 11 and deck/Frobenius checks remain mandatory.

The complete degree-7/9/11 output lists match the exhaustive R45 collector byte for byte before the degree-13 candidate starts. The original degree-13 producer was terminal first. The new degree-13 export also matches exactly: 580 closed points in 58 deck orbits, comprising 7540 rational source points. No sample replaces the full output comparison. All native attempts and proof receipts are indexed in results.json.

Counted scope and proof boundary

The two candidate counts describe different detector inputs. Polynomial construction, GCD, Frobenius and trace splitting have additional costs; no matched complete timing or counted-work panel was measured. This is not an IC speedup claim. Polynomial certificates are checked in-process and are not separately exported. Full-list equality supplies the independent exhaustive-detector control, not an independent scientific replay.

The Lean module proves conditional completeness from a reconstructed product of linear factors and a nonzero leading scalar. Native arithmetic, the GCD implementation and proper geometry are not formalized. Audit outcomes and all failed attempts, if any, remain in results.json.

Next: actual complete divisor evaluation on R45′s new supports, signed Picard classes, nonzero target action and dependence checks. Proper level-77 map, quotient saturation/polarization, theta and node187 remain open. Fixed n=5 carrier; larger residue degree is not a larger-n Koblitz transfer.